

Mixed applications



1 a $h = 86.52$ b $a = 20.37$ c $a = 10.00$ d $y = 2.4$

2 $l = 20.6 \text{ m}$

3 a $x = 149.12^\circ$ b $\theta = 102^\circ$ c $\theta = 146^\circ$

4 a $\theta = 37.22^\circ$ b $d = 3.4 \text{ cm}$

5 $x = 22^\circ$

6 a $AE = 13.8577$ b $BE = 7.6433$ c $AB = 21.50$ d $BD = 6.3$

7 a $73\,500 \text{ m}$ b $h = 41\,101 \text{ m}$

8 $h = 4.4 \text{ m}$

9 a 74 b $d = 28.35$ c $h = 16.66$

10 a $\angle ACB = 102^\circ$ b $b = 226.55 \text{ m}$ c $h = 117 \text{ m}$

11 a $x = 23.88 \text{ m}$ b $k = 25.95 \text{ m}$ c 65.95 m

12 a 21.89° b 123.11° c $AB = 29.2$

13 a 136° b $SW = 234 \text{ km}$ c $h = 102.58 \text{ km}$

14 a 23° b $a = 35.65$ c $h = 20.4$

15

a 3799 km

b 4915.5 km



c $x = 4014.7 \text{ km}$

16 $\theta = 33.56^\circ$

17 $x = 37.79 \text{ km}$

18

a $x = 48^\circ$

b 132°

19

a $\angle ABC = 90^\circ$

b $x = 27^\circ$

c 308°T

20

a $\frac{3}{2}x^2 \text{ cm}^2$

b $\frac{1}{4}$

21

a $\theta = 106.3$

b $x = 36.9$

c $d = 12.0$

22

a $\angle ASC = 90^\circ$

b $AC = 2222 \text{ km}$

c $x = 36^\circ$

d 195°T

23

a $x = 26.73 \text{ km}$

b $b = 20^\circ$

c $\text{N } 34^\circ \text{W}$

24 $x = 41$

25 $y = 2\sqrt{7} \text{ cm}$

26

a 47.13 m

b 76.3 m

c $\theta = 23^\circ$

27

a $\theta = 52.6^\circ$

b 136.66 cm^2